

EFFICIENT COMPUTATION OF TRIPLE SCHUBERT STRUCTURE CONSTANTS

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ABSTRACT.

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1. INTRODUCTION

2. FOUNDATIONAL FORMULAS

2.1. Notation and definitions.

Definition 2.1 ($\mathbf{c}$, dominant permutations, θ , μ). We say that a permutation μ is *dominant* if $\mathbf{c}_i(\mu) \geq \mathbf{c}_{i+1}(\mu)$ for all i . For a permutation $v \in S_\infty$, we define a dominant permutation $\mu(v)$ recursively as follows. If v is already dominant, define $\mu(v) = v$. Otherwise, let i be the maximal index such that $\mathbf{c}_i(v) < \mathbf{c}_{i+1}(v)$, and define

$$\mu(v) = \mu(vs_i)$$

Define $\theta_i(v)$ to be $\mathbf{c}_i(\mu(v))$.

Definition 2.2 ($\xrightarrow{i}$, $\sigma_i(u, v)$, $D_i(u, v)$). Let $u, v \in S_\infty$ and $i \geq 1$ be an integer. We define $u \xrightarrow{i} v$ if there exist distinct integers $b_1, b_2, \dots, b_k$ such that $b_j \neq i$ for all j , $\ell(ut_{i,b_1} \cdots t_{i,b_k}) = \ell(u) + k$ for all k , and

$$v = ut_{i,b_1} \cdots t_{i,b_k}$$

We define

$$\sigma_i(u, v) = \#\{j \mid b_j < i\}$$

and we define

$$D_i(u, v) = \{u(i)\} \cup \{u(j) \mid u(j) \neq v(j)\}$$

2.2. The formula for double Schubert polynomials.

Theorem 2.3 (Formula for double Schubert polynomials). *Suppose $v \in S_\infty$. Then*

$$\mathfrak{S}_v(x; z) = \sum_{(v_0, \dots, v_k)} \prod_{i=1}^k (-1)^{\sigma_i(v_{i-1}, v_i)} E_{\theta_i(v) - \ell(v_{i-1}, v_i), \theta_i(v)}(x; z_{D_i(v_{i-1}, v_i)})$$

Where $v_{i-1} \xrightarrow{i} v_i$ for all i , $v_0 = 1$, and $v_k = v\mu(v^{-1})$.

2.3. The Pieri formula.

Theorem 2.4 (The Pieri formula, [9, Theorem 7.1]). *Suppose $u \in S_\infty$, $k \geq 1$, and $0 \leq p \leq k$. Let x, y, z be sets of variables. Then*

$$\mathfrak{S}_u(x; y) E_{p, k}(x; z) = \sum_{u \xrightarrow{k} w} E_{p - \ell(u, w), k - \ell(u, w)}(y; z_{P_k(u, w)}) \mathfrak{S}_w(x; y)$$

where $P_k(u, w) = \{u(i) \mid i \leq k \text{ and } u(i) = w(i)\}$.

3. THE **pier**i ALGORITHM

Definition 3.1 (*k*-strip cover, admissible *k*-chain, the (p, k) -Pieri problem). Let $w \in S_n$, let $1 \leq k < n$, and let $\beta \in \{1, \dots, n\} \cup \{\infty\}$. A *k-strip cover* of (w, β) is a transposition t_{ab} of positions a, b with $1 \leq a \leq k < b \leq n$ such that

- (1) $w < wt_{ab}$ in Bruhat order; that is, $w(a) < w(b)$ and no c with $a < c < b$ satisfies $w(a) < w(c) < w(b)$; and
- (2) $w(b) < \beta$.

Its *output* is the pair $(wt_{ab}, w(b))$. An *admissible k-chain of length p* from u is a sequence $u = w_0, w_1, \dots, w_p$ carrying bounds $\infty = \beta_0 > \beta_1 > \dots > \beta_p$ in which (w_t, β_t) is the output of a *k-strip cover* of (w_{t-1}, β_{t-1}) for every t . The (p, k) -Pieri problem for u asks to enumerate every admissible *k-chain* from u of length at most p . These chains index the permutations occurring in $e_p(x_1, \dots, x_k) \mathfrak{S}_u$, where e_p is the elementary symmetric polynomial.

Method 3.2 (**pier**i(u, p, k)). Given $u \in S_n$, $1 \leq k < n$, and $0 \leq p \leq k$, the algorithm returns a list T of pairs (w, t) , one for each admissible *k-chain* from u of length $t \leq p$, recording its endpoint w .

- (1) Initialize $T \leftarrow [(u, 0)]$ and a frontier $F \leftarrow [(u, \infty)]$.
- (2) For $t = 1, \dots, p$, form a new frontier $F' \leftarrow []$ and, for each $(w, \beta) \in F$:
 - (a) compute $A \leftarrow \{a \in \{1, \dots, k\} \mid w(a) < \beta\}$;
 - (b) for each $b = k+1, \dots, n$ with $w(b) < \beta$, and each $a \in A$ for which t_{ab} is a *k-strip cover* of (w, β) , append $(wt_{ab}, w(b))$ to F' and (wt_{ab}, t) to T .
 Then set $F \leftarrow F'$.
- (3) Return T .

Lemma 3.3 (Multiplicity-freeness). *For each t , distinct admissible k -chains from u of length t have distinct endpoints; equivalently, the endpoint map is a bijection from admissible k -chains of length t onto the permutations w with $w \succ^k u$ of length $\ell(u) + t$ produced by the Pieri rule for $e_t(x_1, \dots, x_k)$.*

Proof. The strictly decreasing bounds $\beta_0 > \beta_1 > \dots$ force the values moved leftward across position k to be strictly decreasing, which is exactly the condition making the e_t -Pieri expansion multiplicity-free; see [1]. Reconstructing the chain from its endpoint by peeling off these moves in decreasing order of value shows the map is injective, hence bijective onto the Pieri support. $\square$

Proposition 3.4 (Complexity of **pier**i). *Let $M := |T|$ be the number of admissible k -chains from u of length at most p that **pier**i(u, p, k) returns. The algorithm runs in*

$$O(k(n-k)n \cdot M)$$

elementary operations, that is $O(n^3)$ per enumerated permutation. Moreover, by Lemma 3.3,

$$M \leq \#\{w \in S_n \mid w \geq u, \ell(w) \leq \ell(u) + p\}.$$

Proof. Processing one frontier node (w, β) costs $O(k)$ to build A , and then iterates over the $n - k$ choices of b and the at most k elements of A ; for each pair the Bruhat-cover test and the transposition each scan $O(n)$ positions. Thus a node costs $O(k + (n - k)kn) = O(k(n - k)n)$. By Lemma 3.3 the number of nodes ever placed on a frontier equals $\sum_{t=0}^{p-1} N_t$, where N_t is the number of length- t chains; since $\sum_{t=0}^{p-1} N_t \leq \sum_{t=0}^p N_t = M$, the total cost is $O(k(n - k)n \cdot M)$. The bound on M holds because every endpoint satisfies $w \geq u$ and $\ell(w) \leq \ell(u) + p$. $\square$

4. THE **elemrepr** ALGORITHM

We now describe an algorithm **elemrepr** for expressing a double Schubert polynomial as in Theorem 2.3. Its starting point is the structure of a *dominant* double Schubert polynomial. If μ is dominant with $\mathfrak{c}(\mu) = (\lambda_1, \dots, \lambda_m)$, then $\mathfrak{S}_\mu(x; z)$ factors as a product of factorial elementary symmetric polynomials of maximal degree, one per column of the shape,

$$\mathfrak{S}_\mu(x; z) = \prod_{i=1}^m E_{\lambda_i, \lambda_i}(x; z_i),$$

where z_i is the indicated single indeterminate [5]. For a general v one writes $\mathfrak{S}_v(x; z)$ by applying divided differences in the z -variables to the dominant polynomial $\mathfrak{S}_{\mu(v^{-1})}(x; z)$. By the Leibniz formula, this operation may be carried out one column at a time. Applied to a single column this produces a signed descending expansion.

Definition 4.1 (k -strip descent beneath a bound, the down-Pieri problem). Let $w, \mu \in S_n$ and $1 \leq k < n$. Say w' lies *beneath the bound* μ if

$$\text{inv}(w'\mu) = \text{inv}(\mu) - \text{inv}(w'),$$

that is, $\ell(w'\mu) = \ell(\mu) - \ell(w')$. A k -strip descent of w is a Bruhat descent $w \succ wt_{ab}$ realized by a transposition t_{ab} , with $a < b$ and $k \in \{a, b\}$, that moves the value in the pivot position k either leftward (if $b = k$, with sign -1) or rightward (if $a = k$, with sign $+1$), touching only positions still holding their original value. Given a starting label w , the permutation μ , a degree p , and the column k , the *down-Pieri problem* asks for the signed list of those length- t descending k -strip chains from w ($t \leq p$) whose endpoint lies beneath μ in weak order.

Method 4.2 (**kdown**(w, μ, p, k)). The algorithm returns a list D of triples (w', t, s) , one for each length- t descending k -strip chain from w ($t \leq p$) whose endpoint w' lies beneath μ in weak order, with $s \in \{+1, -1\}$ its sign.

- (1) Initialize $D \leftarrow []$; if w is already beneath μ , append $(w, 0, +1)$. Set a frontier $F \leftarrow [(w, +1)]$.
- (2) For $t = 1, \dots, p$, form $F' \leftarrow []$ and, for each $(w'', s) \in F$:
 - (a) (leftward) for each position $a < k$ with $w''(a) = w(a)$ such that $w'' \succ w''t_{a,k}$ is a Bruhat descent, set $w''' \leftarrow w''t_{a,k}$ and append $(w''', -s)$ to F' ; if w''' is beneath μ , append $(w''', t, -s)$ to D ;
 - (b) (rightward) for each position $b > k$ with $w''(b) = w(b)$ such that $w'' \succ w''t_{k,b}$ is a Bruhat descent, set $w''' \leftarrow w''t_{k,b}$ and append $(w''', +s)$ to F' ; if w''' is beneath μ , append $(w''', t, +s)$ to D .
 Then set $F \leftarrow F'$.
- (3) Return D .

Applied without the bound μ , the signed descending expansion produces many extraneous terms that ultimately cancel. The dominant permutation μ is exactly the bound that discards them: only endpoints beneath μ are retained in D . In the assembled v -side (Method 4.3 below) only these retained labels seed the next column, so the bound prevents the number of live paths from blowing up across columns.

Method 4.3 (**elemrepr**(v), the v -side path dictionaries). Let $\theta := \theta(v^{-1})$ with nonzero parts in columns $1, \dots, L$, and let $vmu := v\mu(v^{-1})$. Processing columns from $i = L$ down to $i = 1$, maintain for each column a set of live labels (initialized at column L to the single label vmu). At column i , with $k = i$ and bound $\mu_i := \mu$ of the shape truncated below column i , call **kdown**(w, μ_i, θ_i, k) for each live label w ; record its signed output triples as the transitions out of w , and let their endpoints be the live labels seeding column $i - 1$. The collected transitions form the path dictionaries used by **schubmult**.

Lemma 4.4 (Correctness of the bound filter). *A permutation w' produced by a descending k -strip chain from w is retained by **kdown**(w, μ, p, k) if and only if w' lies beneath μ , i.e. $\text{inv}(w'\mu) = \text{inv}(\mu) - \text{inv}(w')$. In the assembly of Method 4.3 the retained signed labels are exactly the terms of $\mathfrak{S}_v(x; z)$ contributed through column i , so no extraneous (cancelling) label propagates to column $i - 1$.*

Proof. The retention test is the identity $\text{inv}(w'\mu) = \text{inv}(\mu) - \text{inv}(w')$ checked in step (2); this is precisely the condition that $w' \leq \mu$ in weak order, i.e. $\ell(w'\mu) = \ell(\mu) - \ell(w')$. The factorization of the dominant polynomial by column, together with the column-wise decoupling of the z -divided differences, identifies the beneath- μ endpoints at column i with the surviving monomials of $\mathfrak{S}_v(x; z)$ after processing that column; the extraneous chains are exactly those leaving the shape, which the bound discards before they seed column $i - 1$. $\square$

Proposition 4.5 (Complexity of **kdown**/**elemrepr**). *Let $R := |D|$ be the number of retained signed triples and F^* the total number of frontier nodes generated by **kdown**(w, μ, p, k). The algorithm runs in*

$$O(n^2 \cdot F^*)$$

elementary operations, and $R \leq F^$. Consequently **elemrepr**(v), which invokes **kdown** once per live label across the $L \leq \min(n - 1, \ell_v)$ columns and by Lemma 4.4 propagates only retained labels, runs in $O(n^2 \cdot \Phi_v)$ operations, where Φ_v is the total number of frontier nodes generated over all columns.*

Proof. Processing one frontier node scans $O(n)$ candidate positions, each requiring an $O(n)$ Bruhat-descent test and $O(n)$ transposition, so $O(n^2)$ per node; summing over the F^* nodes gives $O(n^2 F^*)$. Each retained triple arises from a distinct frontier node, so $R \leq F^*$. In **elemrepr**, by Lemma 4.4 the labels seeding column $i - 1$ are only the retained (beneath-bound) endpoints of column i , so the live-label count never includes extraneous paths; summing the per-column frontier work gives $O(n^2 \Phi_v)$. $\square$

5. THE **schubmult** ALGORITHM

Method 5.1 (The algorithm **schubmult**). Fix n and let $u, v \in S_n$. The algorithm **schubmult**(u, v) computes the expansion of $\mathfrak{S}_u(x; y) \mathfrak{S}_v(x; z)$ in the Schubert basis as follows. Set $\theta := \theta(v^{-1})$ and let

$$L := \#\{i \mid \theta_i \neq 0\}$$

be the number of nonzero parts.

The arithmetic takes place in the coefficient ring $R := \mathbb{Z}[y; z]$, in which the structure constants live. The algorithm proceeds in L layers and maintains a hash H that maps each pair (w, λ) —an intermediate u -side permutation w together with a v -side label λ —to a coefficient $c(w, \lambda) \in R$. All coefficients are initialized to 0 except for the single key (u, e) , which is initialized to $c(u, e) = 1$. Write $\mu := \mu(v^{-1})$ and $vmu := v\mu$.

Each layer transforms H by one column of the shape. Fix layer i and put $k := \theta_i$, and group the keys of H by their u -side permutation. For each distinct u -side permutation w appearing in some key:

- (1) Call **pieri**(w, θ_i, θ_i) (Method 3.2) *once* to enumerate the u -side covers $w \xrightarrow{\theta_i} w'$; each such cover carries a degree $a \in \{0, 1, \dots, \theta_i\}$, which is number of inversions it adds. The resulting cover list is reused for every label paired with w and is not recomputed per label.
- (2) For each label λ with $c = c(w, \lambda) \neq 0$, read from **elemrepr**(v) (Method 4.3) the v -side transitions out of λ in column i : each is a signed triple (λ', b, s) with endpoint λ' , degree b , and sign $s \in \{+1, -1\}$.
- (3) For every pair with $w \xrightarrow{\theta_i} w'$ of degree a , triple (λ', b, s) with $a + b \leq \theta_i$, accumulate into the new hash the single ring element

$$c(w', \lambda') \ += \ s \cdot c \cdot E_{\theta_i - a - b, \theta_i - a}(y_A; z_B),$$

where $E_{p,m}(y; z)$ denotes the factorial elementary symmetric polynomial of degree p in the first m variables,

$$E_{p,m}(y; z) := \sum_{1 \leq i_1 < \dots < i_p \leq m} \prod_{j=1}^p (y_{i_j} - z_{i_j - j + 1}),$$

evaluated on the y -alphabet $y_A := \{y_{w'(j)} \mid j \leq \theta_i, w(j) = w'(j)\}$ of the $\theta_i - a$ values the cover leaves fixed in columns $1, \dots, \theta_i$, and the z -alphabet z_B where

$$B = \{i\} \cup \{\lambda(j) \mid j \neq i \text{ and } \lambda(j) = \lambda'(j)\}$$

- (4) After all L layers the coefficient $c(w, vmu)$ at each u -side endpoint w is the structure constant of $\mathfrak{S}_w(x; y)$ in $\mathfrak{S}_u(x; y) \mathfrak{S}_v(x; z)$.

Thus each layer performs, for every matched (cover, triple) pair, the construction of one factorial elementary symmetric polynomial $E_{p,m}(y; z)$, followed by one ring multiplication (by the sign and the running coefficient) and one ring addition into the accumulator $c(w', \lambda')$. The polynomial $E_{p,m}(y; z)$ is rebuilt on each pair by the divide-and-conquer recursion of Lemma 5.2: this recursion is not memoized in the y -alphabet—only the z -index selection is cached—so its construction is a nontrivial cost, recorded as E_n in Definition 5.4, and not a shared lookup. Every coefficient $c(w, \lambda)$ is kept in factored form—a product of the factorial elementary symmetric factors $E_{p,m}(y; z)$ accumulated along its paths—and is never multiplied out into the monomial basis, so it asymptotically stays far smaller than its monomial expansion, whose number of terms is exponential in n .

Lemma 5.2 (Divide-and-conquer splitting of $E_{p,m}$). *Let $y = (y_1, \dots, y_m)$ and let $z = (z_r)_{r \in \mathbb{Z}}$ be an alphabet. For an integer c put*

$$E_{p,m}^{(c)}(y; z) := \sum_{1 \leq i_1 < \dots < i_p \leq m} \prod_{j=1}^p (y_{i_j} - z_{i_j - j + 1 + c}),$$

so that $E_{p,m} = E_{p,m}^{(0)}$. Set $t := \lfloor m/2 \rfloor$, $y' := (y_1, \dots, y_t)$, and $y'' := (y_{t+1}, \dots, y_m)$. Then for $0 \leq p \leq m$ and every integer c ,

$$E_{p,m}^{(c)}(y; z) = \sum_{a=0}^p E_{a,t}^{(c)}(y'; z) E_{p-a, m-t}^{(c+t-a)}(y''; z),$$

with the conventions $E_{0,\ell}^{(c)} = 1$ and $E_{a,\ell}^{(c)} = 0$ for $a > \ell$, and the base cases $E_{1,\ell}^{(c)}(y; z) = \sum_{i=1}^{\ell} (y_i - z_{i+c})$ and $E_{\ell,\ell}^{(c)}(y; z) = \prod_{i=1}^{\ell} (y_i - z_{1+c})$.

Proof. Classify each index set $1 \leq i_1 < \dots < i_p \leq m$ in the defining sum of $E_{p,m}^{(c)}$ by the number a of its members lying in $\{1, \dots, t\}$; the remaining $p-a$ lie in $\{t+1, \dots, m\}$. The first a factors $\prod_{j=1}^a (y_{i_j} - z_{i_j-j+1+c})$ range over all size- a subsets of $\{1, \dots, t\}$ and sum to $E_{a,t}^{(c)}(y'; z)$. For the upper part write $i_{a+r} = t + l_r$ with $1 \leq l_1 < \dots < l_{p-a} \leq m-t$; the corresponding factor is

$$y_{t+l_r} - z_{i_{a+r}-(a+r)+1+c} = y_{l_r}'' - z_{l_r-r+1+(c+t-a)},$$

which is the r -th factor of $E_{p-a, m-t}^{(c+t-a)}(y''; z)$. So the upper factors sum to $E_{p-a, m-t}^{(c+t-a)}(y''; z)$; the terms with $a > t$ contribute 0, consistent with the convention. Summing over a gives the identity. $\square$

Lemma 5.3 (Output support bound of $(2n-1)!!$). *Let $w_0 = w_0(n)$ be the longest element of S_n , with $\mathbf{c}(w_0) = (n-1, n-2, \dots, 1, 0)$, and let $w_1 = w_1(n)$ be the permutation with*

$$\mathbf{c}(w_1) = \mathbf{c}(w_0) + \mathbf{c}(w_0) = (2n-2, 2n-4, \dots, 2, 0),$$

a dominant element of S_{2n-1} with $\ell(w_1) = 2\binom{n}{2}$. For any $u, v \in S_n$, every permutation w occurring in the product $\mathfrak{S}_u \cdot \mathfrak{S}_v$, or in any intermediate product formed during its computation, satisfies $w \leq_L w_1$. The number of such permutations is at most $(2n-1)!!$.

Proof. Consider that

$$\mathfrak{S}_{w_1(n)}(x; y_1, z_1, \dots, y_{n-1}, z_{n-1}) = \mathfrak{S}_{w_0(n)}(x; y_1, \dots, y_{n-1}) \mathfrak{S}_{w_0(n)}(x; z_1, \dots, z_{n-1}).$$

Expanding via the Cauchy formula, we obtain

$$\sum_{w \leq_L w_1(n)} \mathfrak{S}_w(x) \mathfrak{S}_{w w_1(n)^{-1}}(-y_1, -z_1, \dots, -y_{n-1}, -z_{n-1}) = \sum_{u,v} \mathfrak{S}_u(x) \mathfrak{S}_v(x) \mathfrak{S}_{u w_0(n)}(-y) \mathfrak{S}_{v w_0(n)}(-z)$$

By linear independence of the Schubert basis, every w occurring in the product $\mathfrak{S}_u(x) \mathfrak{S}_v(x)$ with nonzero coefficient must satisfy $w \leq_L w_1(n)$. The number of permutations w with $w \leq_L w_1(n)$ is the size of the left weak order interval $[e, w_1(n)]_L$, the double factorial

$$(2n-1)!! = 1 \cdot 3 \cdot 5 \cdots (2n-1) = \frac{(2n-1)!}{2^{n-1} (n-1)!}.$$

This counts all outputs, intermediate ones included. $\square$

Definition 5.4 ($\ell_u, \ell_v, P_u, V, B, \Delta, E_n$). For $u, v \in S_n$ write $\ell_u := \ell(u)$ and $\ell_v := \ell(v)$, and set

$$P_u := \#\{w \mid u \leq w \leq w_1 \text{ and } \ell(w) \leq \ell_u + \ell_v\},$$

with $w_1 = w_1(n)$ as in Lemma 5.3. By that lemma every intermediate and output u -side permutation is counted by P_u , and the number of permutations occurring, intermediates included, is at most $(2n-1)!!$; we write $P_u \leq (2n-1)!!$. Let V be the number of distinct v -side labels produced by $\mathbf{elemrepr}(v)$ across all columns; these are permutations, and, as in Lemma 5.3, we take as given that at most $n!$ of them occur, so $V \leq n!$. Write $\theta := \theta(v^{-1})$. The two *per-layer branchings* are

$$B := \max_i \max_w |\mathbf{pieri}(w, \theta_i, \theta_i)|, \quad \Delta := \max_i \max_{\lambda} \#\{\text{kdown triples } (\lambda', c, \varepsilon) \text{ carried by } \lambda \text{ in column } i\} :$$

B is the maximum number of u -side covers generated by a single **pieri** step (Proposition 3.4), and Δ is the maximum number of v -side kdown triples generated from a single label (Proposition 4.5). Each branching enumerates a family of k -strips indexed by (nonempty) subsets of at most n eligible positions, so

$$B \leq 2^n - 1, \quad \Delta \leq 2^n - 1;$$

the bound on B is sharp, attained at $w = w_0$ (where $\mathbf{pieri}(w_0, n-1, n)$ produces exactly $2^n - 1$ covers).

Finally, let $R(p, m)$ denote the number of coefficient-ring operations (additions and multiplications) performed by the unmemoized divide-and-conquer recursion of Lemma 5.2 in evaluating $E_{p,m}^{(c)}(y; z)$; the count is independent of the offset c . With $t := \lfloor m/2 \rfloor$ it satisfies $R(p, m) = 0$ for $p = 0$ or $p > m$, the base values $R(1, \ell) = R(\ell, \ell) = \ell - 1$, and, for $1 < p < m$,

$$R(p, m) = R(p, t) + R(p, m - t) + 1 + \sum_{a=\max(1, p-m+t)}^{\min(p, t)} (R(a, t) + R(p - a, m - t) + 2).$$

Define

$$E_n := \max_{0 \leq p \leq m \leq n-1} R(p, m).$$

Proposition 5.5. $\log_2 E_n = \Theta(\log^2 n)$; equivalently $E_n = 2^{\Theta(\log^2 n)} = n^{\Theta(\log n)}$, so E_n is superpolynomial in n but subexponential.

Proof. Write $C(m) := \max_{0 \leq p \leq m} R(p, m)$, so $E_n = \max_{m \leq n-1} C(m)$. Throughout put $t := \lfloor m/2 \rfloor$, so the two child sizes t and $m - t$ lie in $\{\lfloor m/2 \rfloor, \lceil m/2 \rceil\}$, and let N be the number of admissible indices a in the recurrence, so $N \leq t + 1 \leq m$.

Upper bound. For $1 < p < m$ the recurrence has $2N + 2$ summands of the form $R(\cdot, t)$ or $R(\cdot, m - t)$, each at most $C(\lceil m/2 \rceil)$, together with $2N + 1$ additive constants; hence, using $C(\lceil m/2 \rceil) \geq 1$ for $m \geq 3$,

$$R(p, m) \leq (2N + 2)C(\lceil m/2 \rceil) + (2N + 1) \leq (4m + 3)C(\lceil m/2 \rceil).$$

Since $C(2) = 1$, induction on m gives $C(m) \leq (4m + 3)^{\lceil \log_2 m \rceil}$: the step uses $4\lceil m/2 \rceil + 3 \leq 4m + 3$ and $1 + \lceil \log_2 \lceil m/2 \rceil \rceil = \lceil \log_2 m \rceil$. Therefore

$$\log_2 C(m) \leq \lceil \log_2 m \rceil \log_2(4m + 3) = O(\log^2 m),$$

and so $\log_2 E_n = O(\log^2 n)$.

Lower bound. Let $I(p, m)$ count the internal nodes—calls with $1 < p < m$ —of the unmemoized recursion tree of $R(p, m)$. Each internal node performs at least one ring addition, so $R(p, m) \geq I(p, m)$, and I satisfies, for $1 < p < m$,

$$I(p, m) = 1 + I(p, t) + I(p, m - t) + \sum_{a=\max(1, p-m+t)}^{\min(p, t)} (I(a, t) + I(p - a, m - t)).$$

Put $J(m) := \sum_{p=0}^m I(p, m)$. Dropping all terms of the sum except $I(a, t)$, a fixed value $a^* \in \{1, \dots, t\}$ is an admissible index in the recurrence for $R(p, m)$ precisely when $a^* \leq p \leq a^* + (m - t)$, which holds for at least $\lfloor m/2 \rfloor$ values of p with $1 < p < m$; summing the resulting contribution $I(a^*, t)$ over those p gives

$$J(m) \geq \left\lfloor \frac{m}{2} \right\rfloor \sum_{a^*=1}^t I(a^*, t) = \left\lfloor \frac{m}{2} \right\rfloor J(t).$$

As $J(4) \geq 1$ (since $I(2, 4) \geq 1$), unrolling this inequality down to a fixed constant size yields

$$\log_2 J(m) \geq \sum_{i=0}^{\lceil \log_2 m \rceil - 3} \log_2 \left\lfloor \frac{m}{2^{i+1}} \right\rfloor = \Omega(\log^2 m).$$

Finally $J(m) \leq (m + 1) \max_p I(p, m) \leq (m + 1) C(m)$, so $\log_2 C(m) \geq \log_2 J(m) - \log_2(m + 1) = \Omega(\log^2 m)$, whence $\log_2 E_n \geq \log_2 C(n - 1) = \Omega(\log^2 n)$. $\square$

Theorem 5.6 (Complexity of **schubmult**). Write $L := \min(n - 1, \ell_v)$. The number of coefficient-ring operations performed by **schubmult**(u, v) is

$$O(n^2 \Phi_v + L \cdot P_u \cdot B \cdot n^3 + L \cdot P_u \cdot V \cdot B \cdot \Delta \cdot E_n),$$

where P_u, V, B, Δ, E_n are as in Definition 5.4 and Φ_v is the v -side precomputation cost of Proposition 4.5. The three summands are, respectively, the one-time v -side precomputation, the live **pi** enumeration (run once per distinct u -side permutation per layer), and the matched-pair ring work; the third dominates in the worst case. Writing that bound as $2^{f(n) + O(\log n)}$,

$$f(n) = 2n \log_2 n + (3 - 2 \log_2 e) n + \Theta(\log^2 n),$$

and, when v is dominant, $f(n) = n \log_2 n + (2 - \log_2 e) n + \Theta(\log^2 n)$, where $\log_2 e = 1/\ln 2 \approx 1.4427$.

Remark 5.7. Here B and Δ are the u -side (Pieri) and v -side (kdown-triple) per-layer branchings and E_n is the per-iteration elementary-symmetric construction cost. Each term of $f(n)$ is a multiplicative factor of the running time, so all are retained. The dominant contribution is the product of the two support counts: the at most

$$P_u \leq (2n-1)!! = \frac{(2n-1)!}{2^{n-1}(n-1)!}$$

distinct u -side permutations and the at most $V \leq n!$ distinct v -side labels, whose product telescopes,

$$P_u V \leq (2n-1)!! n! = \frac{(2n)!}{2^n},$$

so together with the branching factor $B\Delta \leq 4^n$ the two counts and branchings contribute $\log_2(P_u V) + 2n = \log_2((2n)!) + n$; the elementary-symmetric construction adds $\log_2 E_n = \Theta(\log^2 n)$. Applying Stirling in the form $\log_2(m!) = m \log_2 m - m \log_2 e + O(\log m)$ gives $\log_2((2n)!) = 2n \log_2 n + 2n - 2n \log_2 e + O(\log n)$, so

$$f(n) = \log_2((2n)!) + n + \Theta(\log^2 n) = 2n \log_2 n + (3 - 2 \log_2 e) n + \Theta(\log^2 n),$$

and the worst-case running time is $2^{f(n)+O(\log n)} = n^{\Theta(n)} = 2^{\Theta(n \log n)}$. The live **pieri** enumeration contributes only n^3 per distinct u -permutation and so lands in the $O(\log n)$ term; it is decoupled from the v -label count V because **pieri**(w, θ_i, θ_i) is run once per permutation w present in a layer, independent of λ .

Proof. By Method 5.1 the algorithm runs in $L = \#\{i \mid \theta_i(v^{-1}) \neq 0\} \leq \min(n-1, \ell_v)$ layers. Write $\mu := \mu(v^{-1})$, so that $\sum_i \theta_i(v^{-1}) = \ell(\mu)$; note that $\ell(\mu) \geq \ell_v$, with equality if and only if v is dominant. The two sides are yoked: at layer i a u -side degree u_i and a v -side degree v_i are paired with $u_i + v_i = \theta_i$, and the v -side descends from $v\mu$ to the identity, so $\sum_i v_i = \ell(v\mu)$. Hence the total degree added on the u -side is

$$\sum_i u_i = \sum_i \theta_i - \sum_i v_i = \ell(\mu) - \ell(v\mu) = \ell_v,$$

using $\ell(v\mu) = \ell(\mu) - \ell(v)$ (as $v^{-1} \leq \mu$ in weak order). Thus every intermediate u -side permutation satisfies $w \geq u$ and $\ell(w) \leq \ell_u + \ell_v$; by Lemma 5.3 it also satisfies $w \leq w_1(n)$, hence lies in S_{2n-1} , so there are at most P_u of them. The v -side labels are permutations obtained from $v\mu$ by descending k -strip transpositions; by Definition 5.4 there are at most $V \leq n!$ distinct ones. The hash at the start of each layer therefore has at most $P_u \cdot V$ keys.

We account for the work in three parts. First, the entire v -side is precomputed once by **elemrepr**(v) (Method 4.3) at cost $O(n^2 \Phi_v)$ ring operations (Proposition 4.5); thereafter each v -side transition is a table lookup carrying no combinatorial work.

Second, the live **pieri** enumeration. Because every u -side cover strictly raises length, the u -side transitions form a graded acyclic graph, so within a fixed layer no permutation recurs; hence **pieri**(w, θ_i, θ_i) is invoked at most once per distinct u -side permutation present in that layer (Method 5.1, step 1). The number of such invocations is

$$\sum_{i=1}^L |\Phi_i^u| \leq L \cdot P_u,$$

where Φ_i^u is the set of distinct u -side permutations live at layer i ; when consecutive layers share a degree, $\theta_i = \theta_{i+1}$, caching **pieri** on (w, θ_i) collapses this to at most P_u invocations, and the safe bound $L \cdot P_u$ costs only the extra factor $L \leq n$. Each invocation costs $O(n^3)$ per enumerated cover (Proposition 3.4) and returns at most B covers, so the total **pieri** cost is $O(L \cdot P_u \cdot B \cdot n^3)$. This enumeration is run once per permutation, not once per key, so it is independent of the v -label count V .

Third, the matched-pair ring work. Write Φ_i for the keys (w, λ) present at layer i . The inner loop matches one **pieri** cover of w against one kdown triple of λ , executed

$$W = \sum_{i=1}^L \sum_{(w, \lambda) \in \Phi_i} \beta_i(w) \delta_i(\lambda)$$

times, where $\beta_i(w) := |\mathbf{pieri}(w, \theta_i, \theta_i)|$ and $\delta_i(\lambda)$ is the number of kdown triples carried by λ in column i . Bounding $\beta_i(w) \leq B$, $\delta_i(\lambda) \leq \Delta$, and $|\Phi_i| \leq P_u \cdot V$ gives $W \leq L \cdot P_u \cdot V \cdot B \cdot \Delta$. Each executed iteration constructs one factorial elementary symmetric polynomial $E_{p,m}(y; z)$ (with $p \leq m \leq \theta_i \leq n-1$) by

the unmemoized divide-and-conquer recursion of Lemma 5.2, a fresh evaluation costing E_n ring operations (Definition 5.4); the ensuing sign multiplication, ring addition, and $O(n)$ bookkeeping are lower-order. Hence the matched-pair cost is $O(W \cdot E_n) = O(L \cdot P_u \cdot V \cdot B \cdot \Delta \cdot E_n)$.

Summing the three parts gives

$$T = O(n^2 \Phi_v + L \cdot P_u \cdot B \cdot n^3 + L \cdot P_u \cdot V \cdot B \cdot \Delta \cdot E_n),$$

the bound of Theorem 5.6. Since $\Phi_v \leq L \cdot V \cdot (2^n - 1)$ —each of the at most $L \cdot V$ kdown calls generates at most $2^{\theta_i} - 1 \leq 2^n - 1$ frontier nodes—and $V \cdot \Delta \cdot E_n \geq n^3$ in the worst case, the third summand dominates there.

For the worst case take base-2 logarithms of the third summand and substitute $L \leq n$, $P_u \leq (2n - 1)!!$ (Lemma 5.3), $V \leq n!$, $B, \Delta \leq 2^n - 1$, and $\log_2 E_n = \Theta(\log^2 n)$ (Proposition 5.5), using $(2n - 1)!! \cdot n! = (2n)!/2^n$ to combine the two support counts with the $+2n$ from $B\Delta$:

$$\log_2 T \leq \log_2((2n)!) + n + \Theta(\log^2 n) + O(\log n),$$

which is $f(n) + O(\log n)$. We retain every summand: the two support counts and the branchings contribute $\log_2((2n)!) + n$, and the elementary-symmetric construction contributes $\log_2 E_n = \Theta(\log^2 n)$; only the live **pieri** cost, at $3\log_2 n$, falls into the $O(\log n)$ slack. By Stirling $\log_2((2n)!) = 2n \log_2 n + 2n - 2n \log_2 e + O(\log n)$, so $f(n) = 2n \log_2 n + (3 - 2\log_2 e)n + \Theta(\log^2 n)$, giving $T = n^{\Theta(n)} = 2^{\Theta(n \log n)}$. If v is dominant then $\theta(v) = \mathbf{c}(v)$, $\ell(\mu) = \ell_v$, every path step carries a $+1$ sign, and each layer's v -side is a single path, so $\Delta = 1$ and only $O(\ell_v)$ labels are live in place of $V = n!$; then $T = O(L \cdot P_u \cdot \ell_v \cdot B \cdot E_n)$ and $\log_2 T \leq \log_2((2n - 1)!!) + n + \Theta(\log^2 n) + O(\log n)$. Since $(2n - 1)!! = (2n)!/(2^n n!)$, Stirling gives $\log_2((2n - 1)!!) = n \log_2 n + (1 - \log_2 e)n + O(\log n)$, so in this case $f(n) = n \log_2 n + (2 - \log_2 e)n + \Theta(\log^2 n)$. $\square$

Corollary 5.8 (Ordinary Schubert polynomials). *For ordinary (single) Schubert polynomials, computing the structure constants of the product $\mathfrak{S}_u(x) \mathfrak{S}_v(x)$ by the single-variable specialization of **schubmult** takes*

$$O(n^2 \Phi_v + L \cdot P_u \cdot B \cdot n^3 + L \cdot P_u \cdot V \cdot B \cdot \Delta)$$

integer operations, with $L = \min(n - 1, \ell_v)$. Writing the bound as $2^{g(n) + O(\log n)}$, the worst case is

$$g(n) = 2n \log_2 n + (3 - 2\log_2 e)n,$$

and, when v is dominant, $g(n) = n \log_2 n + (2 - \log_2 e)n$. These are the bounds of Theorem 5.6 with the term $\Theta(\log^2 n)$ deleted.

Proof. The single-variable algorithm has the same combinatorial skeleton as Method 5.1: the shape $\theta := \theta(v^{-1})$, the u -side Pieri covers, and the v -side path structure with its signed transitions (λ', b, s) are identical, so the three-part accounting of the proof of Theorem 5.6—the one-time v -side precomputation $O(n^2 \Phi_v)$, the live **pieri** enumeration $O(L \cdot P_u \cdot B \cdot n^3)$ run once per distinct u -side permutation, and the matched-pair work over $W \leq L \cdot P_u \cdot V \cdot B \cdot \Delta$ pairs—carries over verbatim. The single difference is the weight attached to a matched (cover, triple) pair: for double Schubert polynomials it is the factorial elementary symmetric polynomial $E_{p,m}(y; z)$, whereas for ordinary Schubert polynomials the corresponding Pieri coefficient is 1, so the pair contributes only the integer sign $s \in \{+1, -1\}$. Each executed iteration therefore performs one integer multiplication by s and one integer addition— $O(1)$ ring operations—so E_n is replaced by $O(1)$ in the third summand. Taking base-2 logarithms of the dominant (third) summand and substituting $L \leq n$, $P_u \leq (2n - 1)!!$ (Lemma 5.3), $V \leq n!$, $B, \Delta \leq 2^n - 1$ (and $\Delta = 1$ in the dominant case), and combining $(2n - 1)!! \cdot n! = (2n)!/2^n$ with the $+2n$ from $B\Delta$, gives $g(n)$, which is $f(n)$ of Theorem 5.6 without the $\Theta(\log^2 n)$ contribution of E_n . $\square$

Remark 5.9 (Comparison with prior complexity results). The complexity of a single structure constant has been studied more than that of an entire product. For Schur polynomials the coefficients $c_{\lambda\mu}^\nu$ are the Littlewood–Richardson coefficients, and computing one of them is $\#P$ -complete [7]. Since these coefficients can be exponentially large, a single one has polynomial bit-length yet cannot be evaluated in time polynomial in the binary sizes of the input and output unless $\text{FP} = \#P$; the same then holds for the full product $s_\lambda s_\mu = \sum_\nu c_{\lambda\mu}^\nu s_\nu$. The Littlewood–Richardson rule instead enumerates skew tableaux, of which there are $\sum_\nu c_{\lambda\mu}^\nu$, and its running time is polynomial in that number, namely in the coefficients written in unary. Deciding whether a given coefficient is positive is, by contrast, in P , through the saturation theorem [4] and the reduction to hive-polytope nonemptiness [3].

For Schubert polynomials the coefficients c_{uv}^w generalize the Littlewood–Richardson coefficients, recovered in the Grassmannian case, so computing one is $\#P$ -hard; whether they lie in $\#P$ is open, no combinatorial rule being known. The vanishing problem $\{c_{uv}^w = 0\}$ was recently placed in the polynomial hierarchy, in *coAM* assuming the generalized Riemann hypothesis, hence in Σ_2^P [8].

Since a single coefficient is already $\#P$ -hard, neither the bound of Theorem 5.6 nor that of Corollary 5.8 is polynomial in the binary output size, and the two are compared instead per combinatorial witness. Each executed iteration of **schubmult** processes one matched (cover, triple) pair. Corollary 5.8 performs $O(1)$ integer operations at each such pair, so in the single case the number of ring operations is proportional to the number of witnesses. Theorem 5.6 performs $E_n = n^{\Theta(\log n)}$ operations at each pair (Proposition 5.5), a quasi-polynomial factor—superpolynomial but subexponential in n —incurred by rebuilding the factorial elementary symmetric weight; the single specialization removes it because that weight collapses to a sign.

6. THE GRASSMANNIAN CASE: SCHUR AND FACTORIAL SCHUR MULTIPLICATION

When both factors are Grassmannian the product specializes to Schur, respectively factorial Schur, multiplication, and the doubled-staircase support of Lemma 5.3 collapses onto a single box. This tightens the counts of Definition 5.4 and, more consequentially, lowers the leading exponent of the running time from $\Theta(n \log n)$ to $O(\min(k, n-k) \log n)$, a genuine function of the two box dimensions k and $n-k$ that vanishes as $k \rightarrow 0$ or $k \rightarrow n$.

Definition 6.1 (*k-Grassmannian permutation*). A permutation $w \in S_n$ is *k-Grassmannian* if its unique descent lies at position k , that is $w(1) < \dots < w(k)$ and $w(k+1) < \dots < w(n)$ with $w(k) > w(k+1)$. To such a w we associate the partition $\lambda(w)$ with parts

$$\lambda(w)_i = w(k-i+1) - (k-i+1), \quad 1 \leq i \leq k,$$

which has at most k parts, each at most $n-k$; that is, $\lambda(w)$ is contained in the $k \times (n-k)$ box. This assignment is a bijection between k -Grassmannian permutations in S_n and partitions in that box.

For a k -Grassmannian u the single Schubert polynomial $\mathfrak{S}_u(x)$ equals the Schur polynomial $s_{\lambda(u)}(x_1, \dots, x_k)$, and the double Schubert polynomial $\mathfrak{S}_u(x; z)$ equals the factorial Schur polynomial $s_{\lambda(u)}(x; z)$. When u and v are both k -Grassmannian the structure constants of $\mathfrak{S}_u \cdot \mathfrak{S}_v$ are therefore the Littlewood–Richardson coefficients $c_{\lambda(u)\lambda(v)}^{\nu}$ in the single case [2] and the factorial (Littlewood–Richardson) polynomials in the double case [6, 9].

Lemma 6.2 (*Grassmannian output support*). *Let $u, v \in S_n$ be k -Grassmannian. Then $\mu(v^{-1})$ is k -Grassmannian, and every permutation occurring in the product $\mathfrak{S}_u \cdot \mathfrak{S}_v$, or in any intermediate product formed during its computation by **schubmult**, is k -Grassmannian with associated partition contained in the $k \times 2(n-k)$ box. Both the u -side permutations and the v -side labels are of this form. The number of such permutations is at most*

$$G_k := \binom{2n-k}{k},$$

the number of partitions in the $k \times 2(n-k)$ box.

Proof. That $\mu(v^{-1})$ is k -Grassmannian, and that every intermediate u -side permutation and v -side label stays k -Grassmannian, is the structural property of the Grassmannian shape, which we take as given, exactly as the doubled-staircase support was taken as given in Lemma 5.3. It remains to bound the associated partitions. Write $\lambda := \lambda(u)$ and $\mu := \lambda(\mu(v^{-1}))$, both contained in the $k \times (n-k)$ box, so $\lambda_1, \mu_1 \leq n-k$. Any partition ν arising in the product or a partial product satisfies $\nu \supseteq \lambda$, has at most k parts, and obeys $\nu_1 \leq \lambda_1 + \mu_1 \leq 2(n-k)$; hence ν is contained in the $k \times 2(n-k)$ box. The number of partitions in a $k \times 2(n-k)$ box is $\binom{k+2(n-k)}{k} = \binom{2n-k}{k} = G_k$. For fixed k this is $\binom{2n-k}{k} = \Theta(n^k)$, a polynomial of degree k in n ; uniformly in k it obeys

$$G_k \leq \sum_{j \geq 0} \binom{2n-j}{j} = F_{2n+1} = \Theta(\varphi^{2n}),$$

the $(2n+1)$ -st Fibonacci number, with $\varphi = (1 + \sqrt{5})/2$ and $\varphi^2 = (3 + \sqrt{5})/2 < 2.619$; hence $G_k = O(\varphi^{2n}) = O(2.619^n)$. Jointly in n and k , Stirling's approximation for binomial coefficients gives the closed form

$$\log_2 G_k = (2n - k) H_2\left(\frac{k}{2n - k}\right) + O(\log n), \quad H_2(x) := -x \log_2 x - (1 - x) \log_2(1 - x),$$

with H_2 the binary entropy function; since $0 \leq H_2 \leq 1$ this recovers $\log_2 G_k \leq 2n - k$. Symmetric bounds on the binomial, $\binom{2n-k}{k} \leq (2(n - k) + 1)^k$ and $\binom{2n-k}{k} \leq (k + 1)^{2(n-k)}$, give the sharper reading

$$\log_2 G_k = O(\min(k, n - k) \log n),$$

which vanishes to $O(\log n)$ as $k \rightarrow 0$ or $k \rightarrow n$ and peaks at $\Theta(n)$ near $k = n/2$: the running time is governed by the two box dimensions k and $n - k$, never by n alone. Finally, every intermediate shape lies in the same $k \times 2(n - k)$ box, so a single Pieri step produces at most G_k distinct covers; hence the per-step branchings obey $B, \Delta \leq G_k$, and the number of layers is at most $n - k$ (the code of v^{-1} has at most $n - k$ nonzero parts). $\square$

Theorem 6.3 (Complexity in the Grassmannian case). *Let $u, v \in S_n$ be k -Grassmannian and set $G_k = \binom{2n-k}{k}$. With $L \leq n - k$ layers and per-step branchings $B, \Delta \leq G_k$ (Lemma 6.2), the number of coefficient-ring operations performed by **schubmult**(u, v) is*

$$O(n^2 \Phi_v + L \cdot G_k \cdot B \cdot n^3 + L \cdot G_k^2 \cdot B \cdot \Delta \cdot E_n),$$

with E_n as in Definition 5.4. Writing this bound as $2^{f_{\text{Gr}}(n, k) + O(\log n)}$, the worst case is the explicit function of n and k

$$f_{\text{Gr}}(n, k) = 4(2n - k) H_2\left(\frac{k}{2n - k}\right) + \Theta(\log^2 n), \quad H_2(x) = -x \log_2 x - (1 - x) \log_2(1 - x),$$

with H_2 the binary entropy function (Lemma 6.2). This exponent is $O(\min(k, n - k) \log n)$: it depends genuinely on both box dimensions k and $n - k$, vanishes to $\Theta(\log^2 n)$ as $k \rightarrow 0$ or $k \rightarrow n$, and is largest, $\Theta(n)$, near $k = n/2$. For fixed k it is $O(k \log n)$, so the G_k -dependent factors are polynomial in n of degree $O(k)$; the leading exponent $\Theta(n \log n)$ of the general Theorem 5.6 never appears.

Proof. The three-part accounting of the proof of Theorem 5.6 applies verbatim; only the two counts and the branchings change. By Lemma 6.2 both the u -side permutations and the v -side labels are k -Grassmannian with partition in the $k \times 2(n - k)$ box, so the u -side count P_u and the v -side label count V are each replaced by G_k : the u -side count G_k confines the distinct u -permutations, and the v -side count G_k replaces the general bound $V \leq n!$, because the labels are now k -Grassmannian rather than ranging over all of S_n . The hash therefore holds at most G_k^2 keys at once. The live **pieri** enumeration runs once per distinct u -permutation per layer, at most $L \cdot G_k$ times, each $O(n^3 \cdot B)$; the matched-pair work is over $W \leq L \cdot G_k^2 \cdot B \cdot \Delta$ pairs, each building one factorial elementary symmetric polynomial at cost E_n ; and the v -side precomputation is $O(n^2 \Phi_v)$. Summing gives the stated bound, whose worst-case term is the third.

For the worst case take base-2 logarithms of the third summand. Every shape lies in the $k \times 2(n - k)$ box, so a single Pieri step produces at most G_k covers and $B, \Delta \leq G_k$; the number of layers L equals the number of nonzero parts of the code of v^{-1} , at most $n - k$; and $\log_2 E_n = \Theta(\log^2 n)$ (Proposition 5.5). Substituting,

$$\log_2 T \leq \log_2(n - k) + 4 \log_2 G_k + \Theta(\log^2 n) = 4 \log_2 G_k + \Theta(\log^2 n) + O(\log n).$$

By Lemma 6.2, $\log_2 G_k = (2n - k) H_2\left(\frac{k}{2n - k}\right) + O(\log n)$ with H_2 the binary entropy function, so

$$f_{\text{Gr}}(n, k) = 4(2n - k) H_2\left(\frac{k}{2n - k}\right) + \Theta(\log^2 n) = O(\min(k, n - k) \log n).$$

The exponent is controlled by the box dimensions k and $n - k$; it collapses to $\Theta(\log^2 n)$ at the extremes $k \rightarrow 0$ and $k \rightarrow n$ and reaches $\Theta(n)$ only near $k = n/2$. $\square$

Remark 6.4. The general bound of Theorem 5.6 has dominant exponent $\log_2((2n)!) - n = \Theta(n \log n)$, contributed by the $P_u \leq (2n - 1)!!$ distinct u -side permutations and the $V \leq n!$ distinct v -side labels. In the Grassmannian case both counts, and every per-step branching, are confined to the $k \times 2(n - k)$ box and hence bounded by $G_k = \binom{2n-k}{k}$ (Lemma 6.2), so the whole exponent collapses to $4 \log_2 G_k = 4(2n - k) H_2\left(\frac{k}{2n - k}\right) +$

$O(\log n)$. This is $O(\min(k, n-k) \log n)$, a genuine function of both box dimensions: it is largest, $\Theta(n)$, at the balanced point $k = n/2$, and shrinks to $\Theta(\log n)$ as $k \rightarrow 0$ or $k \rightarrow n$, mirroring the shrinking of $\text{Gr}(k, n)$ at the extremes. The improvement over the general algorithm is a factor $n^{\Theta(n)} = 2^{\Theta(n \log n)}$ in the worst case, and for fixed k the collapse is total: the count is polynomial in n of degree $O(k)$.

Corollary 6.5 (Schur multiplication and Littlewood–Richardson coefficients). *Let $u, v \in S_n$ be k -Grassmannian. Computing the structure constants of $\mathfrak{S}_u(x) \mathfrak{S}_v(x)$ —the Littlewood–Richardson coefficients $c_{\lambda(u)\lambda(v)}^\nu$ —by the single-variable specialization of **schubmult** takes*

$$O(\min(n-1, \ell_v) \cdot G_k^2 \cdot B \cdot \Delta)$$

integer operations, with $L \leq n-k$ layers and $B, \Delta \leq G_k$. Writing this as $2^{g_{\text{Gr}}(n,k)+O(\log n)}$, the worst case is the explicit function of n and k

$$g_{\text{Gr}}(n, k) = 4(2n-k) H_2\left(\frac{k}{2n-k}\right), \quad H_2(x) = -x \log_2 x - (1-x) \log_2 (1-x),$$

with H_2 the binary entropy function; this is $O(\min(k, n-k) \log n)$, vanishing to $O(\log n)$ at $k \rightarrow 0$ and $k \rightarrow n$. It is the bound of Theorem 6.3 with the term $\Theta(\log^2 n)$ deleted.

Proof. As in Corollary 5.8, the single-variable weight attached to a matched (cover, triple) pair is the integer sign $s \in \{+1, -1\}$ rather than a factorial elementary symmetric polynomial, so each executed iteration performs $O(1)$ integer operations and E_n is replaced by $O(1)$ in the third summand. The three-part bound of Theorem 6.3 then reads $O(n^2 \Phi_v + L \cdot G_k \cdot B \cdot n^3 + L \cdot G_k^2 \cdot B \cdot \Delta)$, and the substitutions $L \leq n-k$, $B, \Delta \leq G_k$ (Lemma 6.2), $\log_2 G_k = (2n-k) H_2\left(\frac{k}{2n-k}\right) + O(\log n)$ give $g_{\text{Gr}}(n, k) = 4(2n-k) H_2\left(\frac{k}{2n-k}\right) = O(\min(k, n-k) \log n)$. In particular the operation count is polynomial in the number G_k of Grassmannian outputs, of degree at most four, consistent with Remark 5.9: an individual coefficient $c_{\lambda(u)\lambda(v)}^\nu$ may be exponentially large in binary, but the algorithm’s arithmetic is counted per output rather than per bit. $\square$

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